

THE PARTIALLY SYMMETRIC RANK OF A PRODUCT OF TWO W-STATES AND OF BIHOMOGENEOUS BINARY MONOMIALS

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ABSTRACT. For $d_1, d_2 \geq 2$ let $W_d = x^{d-1}y \in \text{Sym}^d(\mathbb{C}^2)$ (up to scale) be the W-state. We prove that the partially symmetric rank of $W_{d_1} \otimes W_{d_2}$ equals $2(d_1 + d_2 - 2)$, so the known upper bound of Gałazka and of Canino–Casarotti–Santarsiero is sharp for every pair. This settles the two-factor case of Question 12 of Oneto–Ventura. The lower bound combines two elementary facts: a point set whose Veronese images span $x^p y \otimes u^q v$ lies on no curve of bidegree $(1, q)$ or $(p, 1)$, and a Sylvester-type catalecticant inequality, in the form used by Wang–Seigal, evaluated at this unbalanced pair of bidegrees. The same argument gives $R(x^a y^b \otimes u^c v^d) = (a+1)(c+1) - (a-b)(c-d)$ for all $a \geq b \geq 1, c \geq d \geq 1$, and a lower bound for products of binary forms whose rank exceeds their border rank. The case of three or more factors is not settled.

1. INTRODUCTION

Let $d_1, \dots, d_k \geq 2$. The *partially symmetric rank* $R(T)$ of $T \in \text{Sym}^{d_1}(\mathbb{C}^2) \otimes \dots \otimes \text{Sym}^{d_k}(\mathbb{C}^2)$ is the least s such that

$$T = \sum_{\ell=1}^s v_{1,\ell}^{\otimes d_1} \otimes \dots \otimes v_{k,\ell}^{\otimes d_k}, \quad v_{j,\ell} \in \mathbb{C}^2.$$

For the W-state $W_d = \sum_{j=1}^d e_1^{\otimes(j-1)} \otimes e_2 \otimes e_1^{\otimes(d-j)}$, which corresponds to the binary form $x^{d-1}y$ up to a nonzero scalar, Oneto and Ventura [8] ask for the exact value of $R(W_{d_1} \otimes \dots \otimes W_{d_k})$ (Question 12). Canino, Casarotti and Santarsiero [3] proved

$$R(W_{d_1} \otimes \dots \otimes W_{d_k}) \leq 2^{k-1} (d_1 + \dots + d_k - 2k + 2);$$

for $k = 2$ this bound is due to Gałazka [5, Thm. 1.5(i)] (numbered Thm. 1.8(i) in the published version). Equality was known for $d_1 = 2$ or $d_2 = 2$ [1, Prop. 4.4], [5, Thm. 1.5(ii)], for $(3, 3)$ [1, Prop. 4.2], and for $(4, 4)$ by an unpublished note of Gałazka [6]. In general only $2 \max(d_1, d_2) \leq R \leq 2(d_1 + d_2 - 2)$ was available [5, 2].

Theorem 1.1. *For all integers $d_1, d_2 \geq 2$,*

$$R(W_{d_1} \otimes W_{d_2}) = 2(d_1 + d_2 - 2).$$

Writing $p = d_1 - 1$ and $q = d_2 - 1$, Theorem 1.1 says $R(T_{p,q}) = 2(p + q)$ for $T_{p,q} = x^p y \otimes u^q v$. The lower bound (Section 3) uses no induction, no case analysis of points at infinity and no computer algebra. Section 4 recalls the matching decomposition. Section 5 extends the argument to products of binary forms and determines the rank of all bihomogeneous binary monomials with positive exponents (Corollary 5.4); Gałazka’s thesis [7, Rem. 1.8] lists this problem as open, and his Remark 1.10 there anticipates the value $2(k + m)$ for $x^k y z^m w$ over $\mathbb{R}$. Section 6 explains why the argument does not extend verbatim to three factors.

2. NOTATION

Let $S = \mathbb{C}[\alpha_0, \alpha_1, \beta_0, \beta_1]$ be bigraded with $\deg \alpha_i = (1, 0)$ and $\deg \beta_i = (0, 1)$. It acts on $R = \mathbb{C}[x, y, u, v]$ by differentiation, $\alpha_0 = \partial_x$, $\alpha_1 = \partial_y$, $\beta_0 = \partial_u$, $\beta_1 = \partial_v$; we write $g \circ F$. For $(A, B) \in \mathbb{Z}_{\geq 1}^2$ and a point $P = ((a : b), (c : e)) \in \mathbb{P}^1 \times \mathbb{P}^1$ with a fixed representative put $\nu(P) = (ax + by)^A (cu + ev)^B \in R_{(A, B)}$. For $g \in S_{(i, k)}$ with $(i, k) \leq (A, B)$,

$$(1) \quad g \circ \nu(P) = \frac{A!}{(A-i)!} \frac{B!}{(B-k)!} g(a, b, c, e) (ax + by)^{A-i} (cu + ev)^{B-k}.$$

For a finite set $Z \subset \mathbb{P}^1 \times \mathbb{P}^1$ let $(I_Z)_{(i, k)} \subseteq S_{(i, k)}$ be the forms vanishing on Z and $h_Z(i, k) = \dim S_{(i, k)} - \dim (I_Z)_{(i, k)}$; this is the rank of the evaluation matrix $[g_r(P)]_{P \in Z, r}$ and does not depend on representatives. For $F \in R_{(A, B)}$, $\text{Ann}(F) = \{g \in S : g \circ F = 0\}$ and $\text{Cat}_{(i, k)}(F) : S_{(i, k)} \rightarrow R_{(A-i, B-k)}$, $g \mapsto g \circ F$. A decomposition of length n of F is $F = \sum_{j=1}^n w_j \nu(P_j)$; $R(F)$ is the least such n . We work over $\mathbb{C}$; all arguments use only that the constants in (1) are nonzero.

3. THE LOWER BOUND

Fix $p, q \geq 1$, $A = p + 1$, $B = q + 1$ and $T = x^p y u^q v \in R_{(A, B)}$.

Lemma 3.1. *Let $F = \sum_j w_j \nu(P_j)$ with $Z = \{P_j\}$ (zero weights allowed) and $g \in (I_Z)_{(i, k)}$ with $(i, k) \leq (A, B)$. Then $g \circ F = 0$.*

Proof. Immediate from (1), since $g(P_j) = 0$ for all j . $\square$

Lemma 3.2. *For $0 \leq i \leq 1$ and $0 \leq k \leq q$ we have $\text{Ann}(T)_{(i, k)} = \beta_1^2 S_{(i, k-2)}$ (which is 0 for $k \leq 1$). Symmetrically, $\text{Ann}(T)_{(i, k)} = \alpha_1^2 S_{(i-2, k)}$ for $0 \leq k \leq 1$ and $0 \leq i \leq p$. In particular $\text{rank Cat}_{(1, q)}(T) = 4$.*

Proof. Let $G = \sum_{t=0}^k c_t \beta_0^{k-t} \beta_1^t$ with $k \leq q$. Terms with $t \geq 2$ annihilate $u^q v$, while $\beta_0^k \circ u^q v = \frac{q!}{(q-k)!} u^{q-k} v$ and, for $k \geq 1$, $\beta_0^{k-1} \beta_1 \circ u^q v = \frac{q!}{(q-k+1)!} u^{q-k+1} v$ are nonzero and linearly independent. Hence $G \circ u^q v = 0$ if and only if $\beta_1^2 \mid G$. For $i = 0$, $g \circ T = x^p y \cdot (G \circ u^q v)$. For $i = 1$ write $g = \alpha_0 G_0 + \alpha_1 G_1$; then $g \circ T = p x^{p-1} y (G_0 \circ u^q v) + x^p (G_1 \circ u^q v)$, and since $x^{p-1} y$ and x^p are linearly independent, $g \circ T = 0$ if and only if β_1^2 divides G_0 and G_1 . The symmetric statement is proved in the same way using $q \geq 1$. Finally $\dim S_{(1, q)} = 2(q+1)$ and $\dim \text{Ann}(T)_{(1, q)} = 2(q-1)$, so $\text{rank Cat}_{(1, q)}(T) = 4$; its image is $\langle x^{p-1} y, x^p \rangle \otimes \langle u, v \rangle$. $\square$

Lemma 3.3. *Let $Z \subset \mathbb{P}^1 \times \mathbb{P}^1$ be finite with $T \in \text{span } \nu(Z)$. Then $(I_Z)_{(1, k)} = 0$ for $0 \leq k \leq q$ and $(I_Z)_{(i, 1)} = 0$ for $0 \leq i \leq p$. In particular $h_Z(1, q) = 2(q+1)$ and $h_Z(p, 1) = 2(p+1)$.*

Proof. Suppose $(I_Z)_{(1, k)} \neq 0$ for some $k \leq q$, take k minimal and $0 \neq g \in (I_Z)_{(1, k)}$. By Lemma 3.1, $g \in \text{Ann}(T)$, so by Lemma 3.2 $k \geq 2$ and $g = \beta_1^2 g_1$ with $g_1 \neq 0$. The form $\beta_1 g_1 \in S_{(1, k-1)}$ is nonzero and vanishes on Z : at a point with $e \neq 0$ we have $0 = g(P) = e^2 g_1(P)$, and at a point with $e = 0$ the factor β_1 vanishes. This contradicts the minimality of k . The second statement is symmetric, using α_1 . $\square$

Lemma 3.4 (catalecticant–Sylvester inequality). *Let $F = \sum_{j=1}^n w_j \nu(P_j) \in R_{(A, B)}$ with all $w_j \neq 0$, $Z = \{P_j\}$, and $(i, k) \leq (A, B)$. Then*

$$n \geq h_Z(i, k) + h_Z(A-i, B-k) - \text{rank Cat}_{(i, k)}(F).$$

Proof. Consider the bilinear form $b(g, h) = (gh) \circ F$ on $S_{(i, k)} \times S_{(A-i, B-k)}$. Since $b(g, h) = h \circ (g \circ F)$ and the apolarity pairing $S_{(A-i, B-k)} \times R_{(A-i, B-k)} \rightarrow \mathbb{C}$ is perfect, $\text{rank } b = \text{rank Cat}_{(i, k)}(F)$. By (1), $b(g, h) = A! B! \sum_j w_j g(P_j) h(P_j)$, so the Gram matrix of b is $A! B! V_1^T W V_2$ with $V_1 = [g_r(P_j)]$, $V_2 = [h_s(P_j)]$ and $W = \text{diag}(w_j)$ invertible. Here $\text{rank } V_1 = h_Z(i, k)$ and $\text{rank } V_2 = h_Z(A-i, B-k)$,

and Sylvester's rank inequality $\text{rank}(XY) \geq \text{rank } X + \text{rank } Y - n$ for $X = V_1^\top W$ (n columns) and $Y = V_2$ gives the claim. $\square$

Lemma 3.4 is, up to notation, the key step in the proof of [9, Thm. 1.8 and Thm. 3.13]; it requires nonzero weights.

Proof of the lower bound in Theorem 1.1. Let $T = \sum_{j=1}^n w_j \nu(P_j)$ be a decomposition of minimal length, so all $w_j \neq 0$, and put $Z = \{P_j\}$. Apply Lemma 3.4 with $(i, k) = (1, q)$, whose complement is $(A - 1, B - q) = (p, 1)$. By Lemma 3.3, $h_Z(1, q) = 2q + 2$ and $h_Z(p, 1) = 2p + 2$; by Lemma 3.2, $\text{rank Cat}_{(1,q)}(T) = 4$. Hence $n \geq (2q + 2) + (2p + 2) - 4 = 2(p + q)$. $\square$

4. THE UPPER BOUND

For completeness we recall an explicit decomposition with $2(p + q)$ terms; it is Gałazka's construction [5, Thm. 1.5(i)] and, for $d_i \geq 3$, a case of [3, Thm. 1.1].

Proposition 4.1. *Let $N = p + q$. Then*

$$\sum_{\varepsilon \in \{\pm 1\}} \sum_{a^N = \varepsilon^{q-1}} \varepsilon (ax + y)^{p+1} (\varepsilon a u + v)^{q+1} = 2N(p+1)(q+1) x^p y u^q v,$$

and the $2N$ points $((a : 1), (\varepsilon a : 1))$ are pairwise distinct. Hence $R(T_{p,q}) \leq 2(p + q)$.

Proof. The coefficient of $x^i y^{A-i} u^k v^{B-k}$ is $\binom{A}{i} \binom{B}{k} \sum_{\varepsilon} \varepsilon^{k+1} \sum_a a^t$ with $t = i + k \in [0, N + 2]$. For fixed ε the a run through a coset $\zeta_{\varepsilon} \mu_N$, so $\sum_a a^t = N \zeta_{\varepsilon}^t$ if $N \mid t$ and 0 otherwise. If $t = 0$ the coefficient is $N \sum_{\varepsilon} \varepsilon = 0$. If $t = N$ then $\zeta_{\varepsilon}^N = \varepsilon^{q-1}$ and the coefficient is $\binom{A}{i} \binom{B}{k} N \sum_{\varepsilon} \varepsilon^{k+q}$, which vanishes unless $k \equiv q \pmod{2}$; since $0 \leq i \leq p + 1$ forces $k \in \{q - 1, q, q + 1\}$, only $(i, k) = (p, q)$ survives, with coefficient $2N(p + 1)(q + 1)$. The value $t = 2N$ occurs only for $N = 2$ with $i = k = 2$, where the coefficient is $N \zeta_{\varepsilon}^4 \sum_{\varepsilon} \varepsilon^3 = 0$. For fixed ε the values a are distinct; for q even the two cosets are disjoint, and for q odd the points $((a : 1), (a : 1))$ and $((a : 1), (-a : 1))$ differ. $\square$

This completes the proof of Theorem 1.1. $\square$

5. PRODUCTS OF BINARY FORMS

Let $F \in \mathbb{C}[x, y]_A$ and $G \in \mathbb{C}[u, v]_B$ be nonzero with $A, B \geq 1$. By Sylvester's theorem $\text{Ann}(F) = (\varphi, \psi)$ is a complete intersection with $\deg \varphi = r \leq \deg \psi = s = A + 2 - r$, and $\text{br}(F) = r$; likewise $\text{Ann}(G) = (\varphi', \psi')$ with degrees $r' \leq s' = B + 2 - r'$. Consider the hypothesis

$$(H_F) : \quad r = s, \text{ or } r < s \text{ and } \varphi \text{ has a repeated linear factor.}$$

By Comas-Seiguer [4], (H_F) holds if and only if $R(F) = s = A + 2 - \text{br}(F)$. For the product, $S / \text{Ann}(F \otimes G) \cong S_F \otimes S_G$ with $S_F = \mathbb{C}[\alpha] / \text{Ann}(F)$, $S_G = \mathbb{C}[\beta] / \text{Ann}(G)$; hence

$$(2) \quad \text{Ann}(F \otimes G)_{(i,k)} = \text{Ann}(F)_i \otimes \mathbb{C}[\beta]_k + \mathbb{C}[\alpha]_i \otimes \text{Ann}(G)_k, \quad \text{rank Cat}_{(i,k)}(F \otimes G) = h_F(i) h_G(k).$$

Lemma 5.1. *Assume (H_G) and let Z be finite with $F \otimes G \in \text{span } \nu(Z)$. Then $(I_Z)_{(r-1,k)} = 0$ for all $k \leq s' - 1$. Symmetrically, (H_F) implies $(I_Z)_{(i,r'-1)} = 0$ for all $i \leq s - 1$.*

Proof. For $i \leq r - 1$ and $k \leq s' - 1$ we have $\text{Ann}(F)_i = 0$ and $\text{Ann}(G)_k = \varphi' \mathbb{C}[\beta]_{k-r'}$, so by (2) $\text{Ann}(F \otimes G)_{(i,k)} = \varphi' S_{(i,k-r')}$. Take $k \leq s' - 1$ minimal with $(I_Z)_{(r-1,k)} \neq 0$ and $0 \neq g$ in it; note $(r - 1, k) \leq (A, B)$. By Lemma 3.1, $g \in \text{Ann}(F \otimes G)$, hence $k \geq r'$, $r' < s'$ and $\varphi' \mid g$. By (H_G) , $\varphi' = \lambda^2 \mu$ with $\lambda \in \mathbb{C}[\beta]_1$, so $g = \lambda^2 g_1$, and λg_1 is a nonzero form of bidegree $(r - 1, k - 1)$ vanishing on Z , contradicting minimality. $\square$

Theorem 5.2. *Under (H_F) and (H_G) ,*

$$R(F \otimes G) \geq rs' + sr' - rr' = R(F)R(G) - (R(F) - \text{br}(F))(R(G) - \text{br}(G)).$$

Proof. Let Z carry a decomposition of minimal length. By Lemma 5.1, $h_Z(r-1, s'-1) = rs'$ and $h_Z(s-1, r'-1) = sr'$; the bidegrees $(r-1, s'-1)$ and $(s-1, r'-1)$ are complementary in (A, B) . By (2), $\text{rank Cat}_{(r-1, s'-1)}(F \otimes G) = h_F(r-1)h_G(s'-1) = r r'$. Lemma 3.4 gives the bound. $\square$

Proposition 5.3. *Under (H_F) alone, $R(F \otimes G) \geq s r' = R(F) \text{br}(G)$ for every nonzero G . Consequently $R(F \otimes G) = R(F)R(G)$ whenever $R(F) = \text{br}(F)$ or $R(G) = \text{br}(G)$.*

Proof. By Lemma 5.1, $|Z| \geq h_Z(s-1, r'-1) = sr'$. If $R(G) = \text{br}(G) = r'$ and (H_F) holds this gives $R(F)R(G)$; if $R(F) = r$ and $R(G) = r'$ the flattening bound rr' suffices; the remaining case follows by symmetry, and $R(F \otimes G) \leq R(F)R(G)$ always. $\square$

Corollary 5.4. *For all integers $a \geq b \geq 1$ and $c \geq d \geq 1$,*

$$R(x^a y^b \otimes u^c v^d) = (a+1)(c+1) - (a-b)(c-d).$$

Proof. For $F = x^a y^b$ we have $\text{Ann}(F) = (\alpha_1^{b+1}, \alpha_0^{a+1})$, so $r = b+1$, $s = a+1$ and (H_F) holds; similarly for $G = u^c v^d$. Theorem 5.2 gives $R \geq (b+1)(c+1) + (a+1)(d+1) - (b+1)(d+1) = (a+1)(c+1) - (a-b)(c-d)$, which coincides with Gałazka's upper bound $(k_0+1)(l_1+1) + (k_1+1)(l_0+1) - (k_1+1)(l_1+1)$ [5, Thm. 1.5(i)] for $(k_0, k_1, l_0, l_1) = (a, b, c, d)$. $\square$

Theorem 1.1 is the case $b = d = 1$. Previously the value was known for $a = b$ or $c = d$ [5, Thm. 1.5(ii)], for $(a, c) = (b+1, d+1)$ [5, Thm. 1.5(i),(iii)], and for $x^3 y u^3 v$ [6].

Remark 5.5. The hypotheses cannot be dropped. For $(x^2 y + y^3) \otimes uv$ the form $\varphi = \alpha_1^2 - 3\alpha_0^2$ is squarefree, the bound of Theorem 5.2 would be 6, and $R = 4$.

6. THREE OR MORE FACTORS

For $T = x^p y \otimes u^q v \otimes s^r t$ with $p, q, r \geq 2$, every complementary pair of multidegrees contains a member with at least two entries ≥ 2 , and in such multidegrees $\text{Ann}(T)$ contains squarefree forms such as $(\alpha_1 \beta_0 - \alpha_0 \beta_1)(\alpha_1 \beta_0 + \alpha_0 \beta_1)$. Lemma 3.3 then fails and the argument does not give $2^{k-1}(d_1 + \dots + d_k - 2k + 2)$. The case $k \geq 3$ of Question 12, for instance $R(W_3 \otimes W_3 \otimes W_3)$ with upper bound 20, remains open.

7. COMPUTATIONAL CHECKS AND PROVENANCE

The proofs above are complete and do not rely on computation. As independent sanity checks we verified exactly over $\mathbb{Q}$ Lemma 3.2, the ranks $\text{rank Cat}_{(1,q)}(T) = \text{rank Cat}_{(p,1)}(T) = 4$ and Lemma 3.4 on 504 random decompositions $(p, q \leq 7)$; over finite fields $\mathbb{F}_P$ with P larger than the degrees, that no curve of bidegree $(1, q)$ or $(p, 1)$ supports T (about 10^6 forms), that no set of $2(p+q) - 1$ rational points spans T for small (p, q) , and that no 8 points of $\mathbb{P}^1 \times \mathbb{P}^1(\mathbb{F}_5)$ span $x^2 y^2 \otimes u^2 v$ (30 260 340 subsets); and Proposition 4.1 together with equality in Lemma 3.4 for ten pairs (p, q) up to $(7, 4)$. Scripts and logs accompany this note.

Provenance. This work was carried out in the context of the OpenTorus project (<https://github.com/maximilianbehr/OpenTorus>), an open-source AI agent for open mathematical problems; an OpenTorus campaign on this catalog entry produced the first apolarity-based sketch for the case $(3, 4)$, and the proof given here was developed from it.

AI disclosure. The results and this manuscript were produced with extensive assistance from AI agents (Anthropic Claude). The arguments were checked by two independent AI-agent reviews and by re-deriving every step; they have not been reviewed by a human referee, and no formal (Lean) verification was performed. The literature search for prior results was bounded.

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